

IMPERIAL BEACH NOLF, CA, USA

WMO: 722909

Lat: 32.567N

Lon: 117.117W

Elev: 24

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 93115

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	42.3	44.1	14.7	11.6	62.7	20.5	15.4	59.4	24.4	58.5	20.9	59.2	6.0	90	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	85.9	64.5	81.4	65.7	78.7	65.6	71.8	79.8	70.4	77.7	69.3	76.0	8.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.9	106.1	75.7	67.6	101.6	75.2	66.4	97.5	74.1	35.5	80.1	34.4	77.8	33.4	76.2	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.5	14.8	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.5	57.1	56.5	57.8	59.8	62.2	65.5	68.8	69.8	68.9	65.3	61.0	56.1
	DBStd	6.41	4.12	4.50	4.05	3.57	3.46	3.28	3.14	4.49	5.48	4.61	4.21	3.92
	HDD50	5	2	1	0	0	0	0	0	0	0	0	0	2
	HDD65	1496	248	241	227	162	101	30	5	6	16	50	134	276
	CDD50	4550	221	184	243	294	378	464	584	615	568	476	331	192
	CDD65	566	2	4	5	6	14	44	124	156	134	61	15	1
	CDH74	2370	45	62	67	72	103	68	191	541	619	394	190	18
	CDH80	570	3	15	19	18	35	8	16	104	197	95	59	1
Wind	WSAvg	6.6	6.5	7.0	6.7	7.2	7.4	7.0	6.7	6.4	6.1	5.8	6.2	6.8
Precipitation	PrecAvg	7.5	1.4	0.9	1.5	0.6	0.2	0.1	0.0	0.0	0.1	0.2	1.4	1.1
	PrecMax	16.7	5.8	3.1	5.4	3.2	1.1	0.5	0.1	0.1	1.5	0.8	6.4	4.4
	PrecMin	4.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.1	1.3	0.9	1.3	0.8	0.2	0.2	0.0	0.0	0.3	0.2	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.3	81.9	81.5	82.4	85.4	80.3	81.4	87.2	95.3	88.3	87.7	76.6
		MCWB	53.9	55.7	57.9	57.7	60.1	68.0	69.9	71.3	72.0	63.3	57.5	51.4
	2%	DB	73.8	74.0	73.4	74.5	75.3	75.3	78.1	81.5	84.6	82.2	78.7	71.1
		MCWB	52.9	54.4	56.8	57.8	60.2	65.0	69.1	70.2	68.7	63.0	56.9	53.7
	5%	DB	69.7	68.9	68.3	70.2	70.2	72.9	76.0	78.6	79.5	78.1	74.2	67.2
		MCWB	53.3	55.6	57.6	58.7	60.7	64.1	67.7	68.9	70.0	64.6	56.5	54.6
	10%	DB	65.9	65.5	65.7	67.5	68.0	71.1	74.3	76.6	76.8	74.9	70.2	64.8
		MCWB	54.5	55.9	57.2	58.7	59.8	63.1	66.6	68.1	68.9	64.4	57.6	54.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	62.0	62.7	63.6	65.4	68.3	72.0	73.3	74.1	71.8	64.4	61.7
		MCDB	64.9	66.9	69.3	71.3	73.2	78.0	80.1	83.1	86.7	79.3	70.7	65.6
	2%	WB	59.5	59.9	60.9	61.4	63.1	66.4	70.0	71.1	72.3	69.2	62.5	60.0
		MCDB	64.4	66.0	68.4	68.6	69.9	73.9	76.5	79.1	80.6	76.6	68.9	64.5
	5%	WB	58.4	58.5	59.5	60.3	61.6	65.2	68.8	69.7	70.9	67.3	61.4	58.7
		MCDB	63.4	64.9	65.4	67.4	68.3	71.5	74.7	77.2	78.3	75.4	68.4	63.4
	10%	WB	57.2	57.2	58.3	59.3	60.3	64.0	67.6	68.8	69.6	65.5	60.1	57.1
		MCDB	62.4	63.6	63.9	66.4	66.6	70.1	73.1	75.7	75.9	72.7	67.3	62.3
Mean Daily Temperature Range	5% DB	MDBR	17.8	16.9	15.0	14.8	11.5	9.4	9.4	10.5	13.5	16.7	18.8	17.8
		MCDBR	25.8	25.6	21.1	22.9	16.3	11.8	11.6	13.1	17.2	23.5	27.2	23.0
		MCWBR	13.6	14.2	12.3	13.2	8.8	6.5	5.9	6.3	8.1	13.4	15.3	14.4
	5% WB	MCDBR	16.0	19.6	16.3	17.8	14.1	11.7	11.7	12.8	14.9	18.8	19.2	16.1
		MCWBR	11.0	12.6	10.5	10.7	8.0	6.6	6.0	6.5	7.4	11.7	13.1	11.5
Clear-Sky Solar Irradiance	taub	0.315	0.327	0.339	0.352	0.372	0.373	0.400	0.399	0.366	0.368	0.324	0.326	
	taud	2.556	2.537	2.486	2.453	2.417	2.417	2.424	2.406	2.477	2.448	2.557	2.552	
	Ebn at Noon	289	296	298	297	290	288	280	279	286	276	283	278	
	Edh at Noon	26	29	33	36	37	37	37	37	33	31	26	25	
All-Sky Solar Radiation	RadAvg	1055	1329	1734	2090	2176	2253	2201	2130	1853	1455	1141	956	
	RadStd	64	108	98	116	169	143	155	102	108	108	78	67	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	+1.17	N/A	-2.83	+1.40	N/A	N/A	N/A	N/A	-189	+414	+220

Nomenclature: See separate page